AI Lab
Week 1

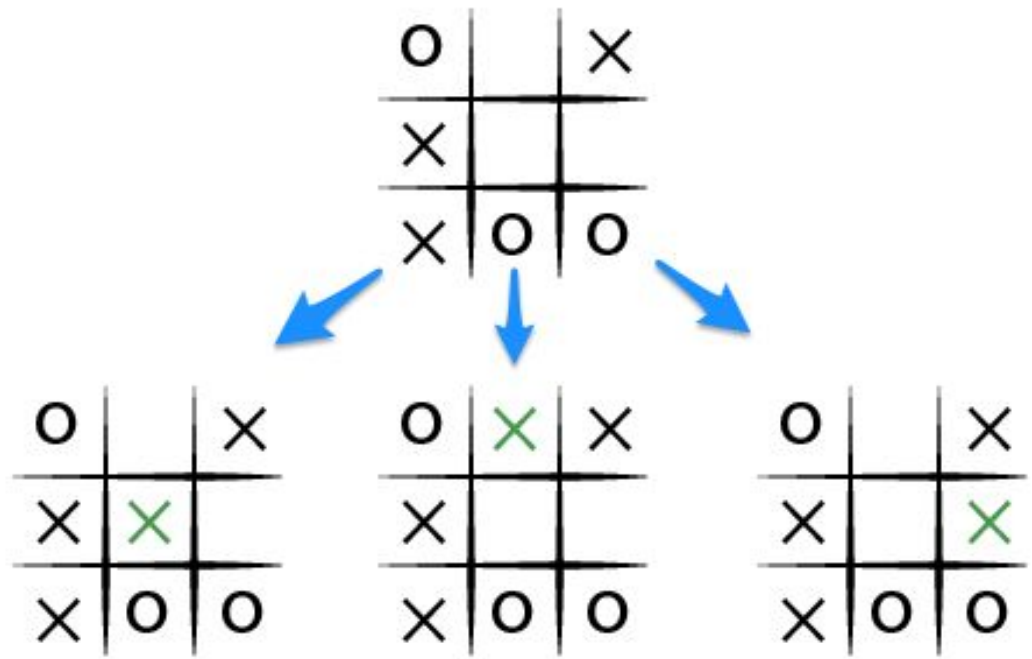
Implement Tic –Tac –Toe Game

Implement vacuum cleaner agent

Tic-Tac-Toe game

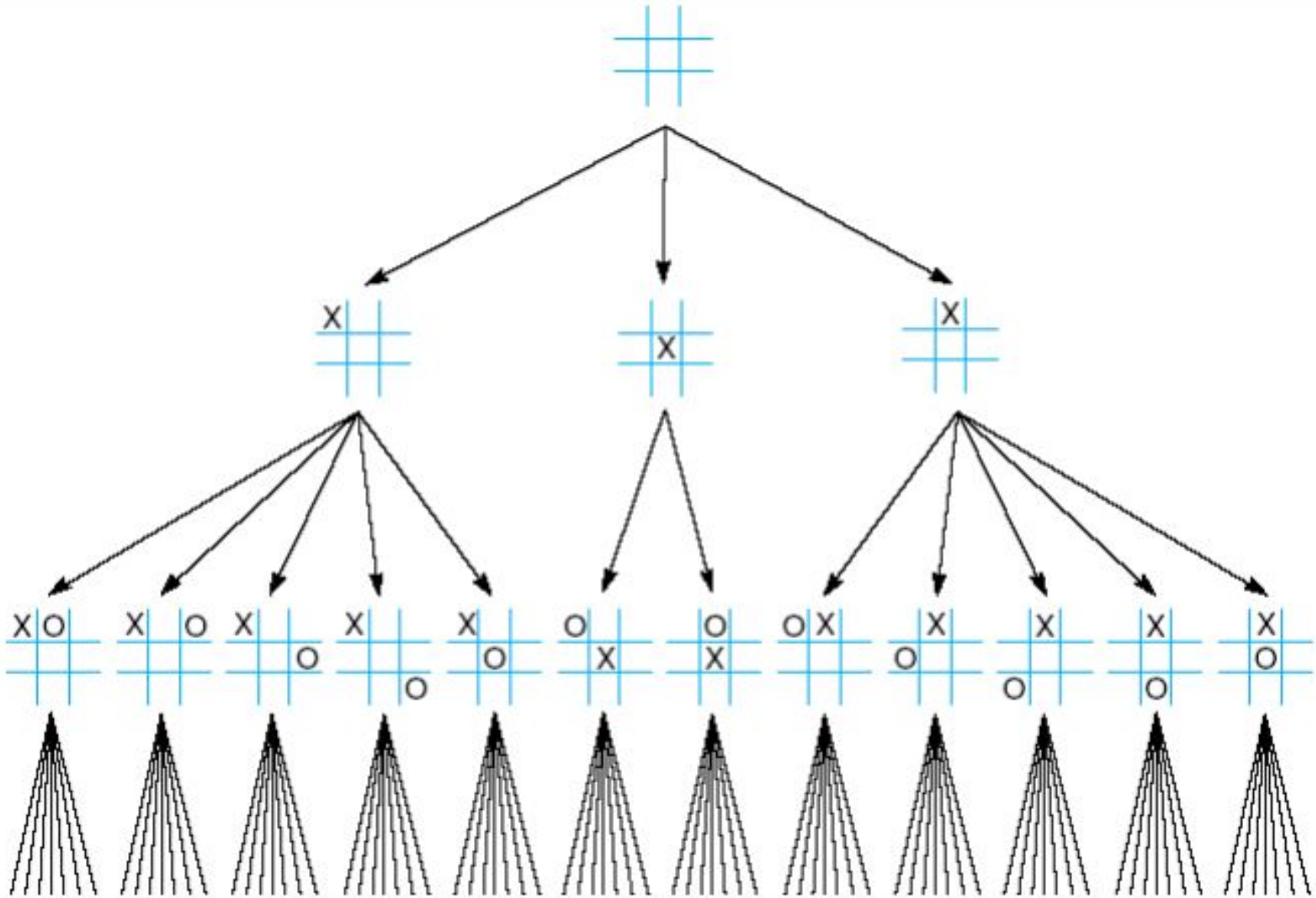
Rules of the Game

- The game is to be played between two people (in this program between HUMAN and COMPUTER).
- One of the player chooses ‘O’ and the other ‘X’ to mark their respective cells.
- The game starts with one of the players and the game ends when one of the players has one whole row/ column/ diagonal filled with his/her respective character (‘O’ or ‘X’).
- If no one wins, then the game is said to be draw.


O X
X
X O O
O X
X X
X O O
O X
X
X O O
O X X
X
X O O
Win! +10

Tic-Tac-Toe game

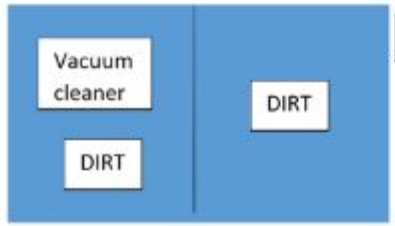
- To solve games using AI, we will introduce the concept of a game tree
- The different states of the game are represented by nodes in the game tree, very similar to the above planning problems.
- .In the game tree, the nodes are arranged in levels that correspond to each player's turns in the game so that the “root” node of the tree (usually depicted at the top of the diagram) is the beginning position in the game.
- In tic-tac-toe, this would be the empty grid with no Xs or Os played yet. Under root, on the second level, there are the possible states that can result from the first player’s moves, be it X or O. We call these nodes the “children” of the root node.
- Each node on the second level, would further have as its children nodes the states that can be reached from it by the opposing player's moves. This is continued, level by level, until reaching states where the game is over. In tic-tac-toe, this means that either one of the players gets a line of three and wins, or the board is full and the game ends in a tie.



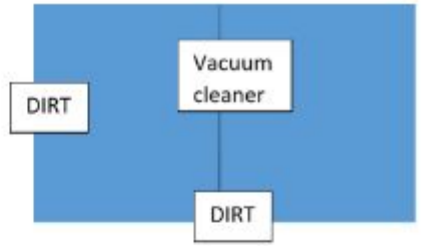
Implement Tic –Tac –Toe Game
Implement vacuum cleaner agent

- Vacuum cleaner problem is a well-known search problem for an agent which works on Artificial Intelligence.
- In this problem, our vacuum cleaner is our agent. It is a goal based agent, and the goal of this agent, which is the vacuum cleaner, is to clean up the whole area.
- So, in the classical vacuum cleaner problem, we have two rooms and one vacuum cleaner. There is dirt in both the rooms and it is to be cleaned. The vacuum cleaner is present in any one of these rooms.
- So, we have to reach a state in which both the rooms are clean and are dust free.
- So, there are eight possible states possible in our vacuum cleaner problem. These can be well illustrated with the help of the following diagrams:

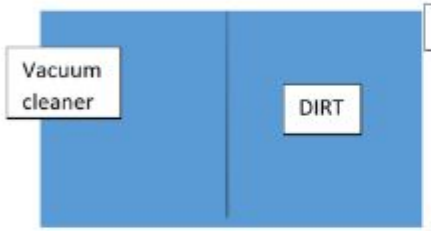
1


Vacuum cleaner
DIRT
DIRT

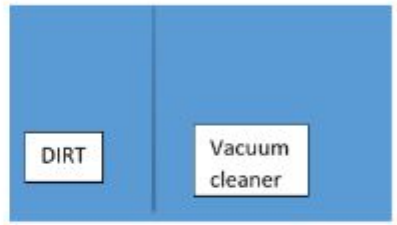
2


Vacuum cleaner
DIRT
DIRT

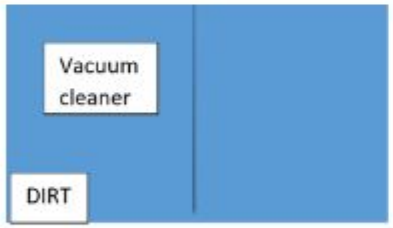
3


Vacuum cleaner
DIRT

4


DIRT
Vacuum cleaner

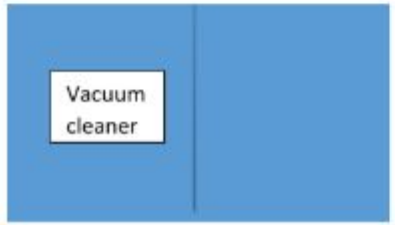
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Vacuum cleaner
DIRT

6


Vacuum cleaner
DIRT

7


Vacuum cleaner

8


Vacuum cleaner